

Predicting consumer centiment based on university canteen data

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Abstract:

This project investigates the key factors influencing customer satisfaction at a university canteen using transactional data collected over 25 days in April 2024. The dataset comprises 200 purchase records including item type, price, quantity, payment method, weather conditions, and special offer availability. The primary objective is to develop predictive models that identify which variables significantly impact customer satisfaction scores (rated 3-5). Exploratory data analysis was conducted to visualize relationships between satisfaction and potential predictors such as item category, weather, and total spending. Two modeling approaches were implemented and compared: multiple linear regression treating satisfaction as a continuous variable, and ordinal logistic regression recognizing the ordered nature of the satisfaction ratings. Model diagnostics, including residual analysis and multicollinearity checks, were performed to validate assumptions. Results indicate that special offers, item type, and weather conditions significantly influence customer satisfaction, while payment method shows minimal effect. The findings provide actionable insights for canteen management to enhance customer experience through targeted interventions. This work demonstrates the application of regression techniques to a real-world community engagement problem.

Introduction

• Project Background and Relevance

Customer satisfaction is a critical metric for any service-oriented establishment, including educational institution canteens that serve students and staff daily. Understanding what drives customer satisfaction enables management to make data-informed decisions that enhance the dining experience, leading to increased customer loyalty, positive word-of-mouth, and ultimately, improved business performance. In the context of a university campus, where students spend a significant portion of their time, canteen satisfaction can even impact overall student well-being and campus life quality.

This project leverages transactional data from a university canteen to identify the key determinants of customer satisfaction. By analyzing variables such as item type, spending amount, weather conditions, payment methods, and special offer availability, we aim to uncover patterns and relationships that explain why some transactions result in higher satisfaction scores than others. The findings will provide actionable insights for canteen operators to optimize their services, menu offerings, and operational strategies.

• Technology & Tools used

The project utilizes the following technologies and tools:

R Programming Language: Primary tool for data analysis, statistical modeling, and visualization.

R Libraries:

- tidyverse for data manipulation and ggplot2 for visualization
- MASS for ordinal logistic regression
- car for regression diagnostics and VIF calculation
- effects for visualizing model effects

- corrrplot for correlation analysis

RStudio: Integrated Development Environment for code development and execution

Microsoft Excel: Initial data inspection and validation

• Purpose of the Project

The primary purposes of this project are:

- To demonstrate the application of statistical regression techniques to a real-world dataset
- To identify which factors significantly influence customer satisfaction at a university canteen
- To provide evidence-based recommendations for improving canteen services
- To fulfill the community engagement requirement by addressing a locally relevant problem
- To develop proficiency in R programming for data analysis and statistical modeling
- To document the complete data analysis workflow in a structured report

Objective

• Overview

The primary focus of this project is to identify and quantify the factors that influence customer satisfaction at a university canteen using transactional data. By applying statistical regression techniques learned, this project aims to transform raw transaction records into actionable insights that can help canteen management enhance the dining experience for students and staff.

• Specific Objectives

The following are the specific objectives of this project:

- ✓ **To identify the key determinants of customer satisfaction** by analyzing the relationship between satisfaction scores and potential predictor variables including item type, total spending, payment method, weather conditions, special offer availability, and time-based factors (peak hour, day of week).
- ✓ **To quantify the impact of each significant factor** on customer satisfaction by estimating regression coefficients that indicate the direction and magnitude of influence, enabling canteen management to prioritize improvement efforts where they will have the greatest effect.
- ✓ **To compare two modeling approaches**—multiple linear regression and ordinal logistic regression—for predicting customer satisfaction, evaluating their respective strengths, assumptions, and interpretability in the context of ordered categorical outcome data.
- ✓ **To assess and validate regression model assumptions** including linearity, normality of residuals, homoscedasticity, and absence of multicollinearity, ensuring the reliability and validity of the findings.
- ✓ **To derive evidence-based recommendations** for canteen operations, such as optimal pricing strategies, menu adjustments, targeted promotional offers, and operational improvements based on weather or time-of-day patterns.

- ✓ **To demonstrate proficiency in R programming** for end-to-end data analysis, including data cleaning, feature engineering, exploratory visualization, statistical modeling, and results interpretation.
- ✓ **To fulfill the community engagement requirement** by addressing a locally relevant problem that directly impacts the daily lives of students and staff, and to communicate findings in a format accessible to non-technical stakeholders (canteen management).

• Hypothesis Testing

Based on the objectives, the following hypotheses will be tested in this project:

Null Hypothesis (H_0): There is no significant relationship between customer satisfaction and the predictor variables (item type, total spending, payment method, weather, special offers, peak hour).

Alternative Hypothesis (H_1): At least one predictor variable has a significant relationship with customer satisfaction.

Additionally, specific hypotheses for individual predictors include:

- **H_0 for Special Offers:** Transactions with special offers have the same average satisfaction as transactions without special offers.
 H_1 for Special Offers: Transactions with special offers have significantly different average satisfaction compared to those without.
- **H_0 for Weather:** Customer satisfaction is the same across all weather conditions (Sunny, Cloudy, Rainy).
 H_1 for Weather: Customer satisfaction differs significantly across weather conditions.
- **H_0 for Item Type:** Satisfaction scores are equal across all food items (Sandwich, Drink, Salad, Burger).
 H_1 for Item Type: At least one item type has significantly different satisfaction scores.
- **H_0 for Total Spending:** There is no linear relationship between total spending amount and customer satisfaction.
 H_1 for Total Spending: There is a significant linear relationship between total spending and customer satisfaction.

These hypotheses will be tested at the 5% significance level ($\alpha = 0.05$) using appropriate statistical tests derived from the regression models.

• Target Population

This project is based on a **sample survey** of canteen transactions. The target population for this study is:

All customers (primarily students, faculty, and staff) making purchases at the university canteen during regular operating hours.

The dataset represents transactions occurring between April 1-25, 2024, capturing a typical operational period without major holidays or special events. While the data includes transaction-level records rather than individual customer surveys, each transaction can be considered a snapshot of customer experience at a specific point in time.

The findings and recommendations derived from this analysis are intended to be applicable to:

- The same canteen during similar operational periods
- Other canteens or food service outlets within the university campus

- Potentially similar food service establishments in educational institutions with comparable customer demographics

• Scope and Limitations

Scope:

- ✓ Analysis is confined to variables available in the transactional dataset
- ✓ Focus on quantitative analysis of satisfaction determinants
- ✓ Both descriptive and inferential statistical methods applied

Limitations:

- Satisfaction is measured on a limited 3-point scale (3-5), which may not capture subtle variations in customer experience
- The dataset lacks certain potentially important variables such as food taste ratings, service speed, staff behavior, and cleanliness
- Transaction-level data does not track individual customers over time, so repeat purchase patterns cannot be analyzed and also there is only 25days data available
- Causal relationships cannot be definitively established due to the observational nature of the data

Methodology

Overview

This project follows a structured data analysis pipeline encompassing data collection, preprocessing, exploratory analysis, statistical modeling, and interpretation. The methodology is designed to systematically identify factors influencing customer satisfaction at the university canteen using transactional data.

Data Collection

The dataset `canteen_shop_data.csv` is collected from Kaggle(<https://www.kaggle.com/code/sonialikhan/canteen-shop-analysis>) . All transactions occurring at the canteen during regular operating hours (12:00 PM to 2:00 PM) between April 1-25, 2024 and every transaction during the specified period was recorded, resulting in a complete census of transactions during the observation window. The sample size is 200 transactions (approximately 8 transactions per day on average). Main target is All customers (students, faculty, and staff) making purchases at the university canteen.

Data

11-04-2024	13:00	Sandwich	4.5	1	4.5	85	Cash	185	3	Cloudy	No
11-04-2024	13:15	Drink	1.5	1	1.5	86	Card	186	5	Rainy	Yes
11-04-2024	13:30	Salad	5	2	10	87	Mobile Payment	187	4	Sunny	No
11-04-2024	13:45	Burger	5.5	1	5.5	88	Cash	188	3	Cloudy	Yes
12-04-2024	12:00	Sandwich	4.5	2	9	89	Card	189	4	Sunny	No
12-04-2024	12:15	Drink	1.5	1	1.5	90	Mobile Payment	190	3	Rainy	Yes
12-04-2024	12:30	Salad	5	1	5	91	Cash	191	5	Cloudy	No
12-04-2024	12:45	Burger	5.5	2	11	92	Card	192	4	Sunny	Yes
12-04-2024	13:00	Sandwich	4.5	1	4.5	93	Mobile Payment	193	3	Cloudy	No
12-04-2024	13:15	Drink	1.5	1	1.5	94	Cash	194	5	Rainy	Yes
12-04-2024	13:30	Salad	5	2	10	95	Card	195	4	Sunny	No
12-04-2024	13:45	Burger	5.5	1	5.5	96	Mobile Payment	196	3	Cloudy	Yes
13-04-2024	12:00	Sandwich	4.5	2	9	97	Cash	197	4	Sunny	No
13-04-2024	12:15	Drink	1.5	1	1.5	98	Card	198	3	Rainy	Yes
13-04-2024	12:30	Salad	5	1	5	99	Mobile Payment	199	5	Cloudy	No
13-04-2024	12:45	Burger	5.5	2	11	100	Card	200	4	Sunny	Yes
13-04-2024	13:00	Sandwich	4.5	1	4.5	101	Mobile Payment	201	3	Cloudy	No
13-04-2024	13:15	Drink	1.5	1	1.5	102	Cash	202	5	Rainy	Yes
13-04-2024	13:30	Salad	5	2	10	103	Card	203	4	Sunny	No
13-04-2024	13:45	Burger	5.5	1	5.5	104	Mobile Payment	204	3	Cloudy	Yes
14-04-2024	12:00	Sandwich	4.5	2	9	105	Mobile Payment	205	4	Sunny	No
14-04-2024	12:15	Drink	1.5	1	1.5	106	Cash	206	3	Cloudy	Yes
14-04-2024	12:30	Salad	5	1	5	107	Card	207	5	Rainy	No
14-04-2024	12:45	Burger	5.5	2	11	108	Mobile Payment	208	4	Sunny	Yes
14-04-2024	13:00	Sandwich	4.5	1	4.5	109	Cash	209	3	Cloudy	No
14-04-2024	13:15	Drink	1.5	1	1.5	110	Card	210	5	Rainy	Yes
14-04-2024	13:30	Salad	5	2	10	111	Mobile Payment	211	4	Sunny	No
14-04-2024	13:45	Burger	5.5	1	5.5	112	Cash	212	3	Cloudy	Yes
15-04-2024	12:00	Sandwich	4.5	2	9	113	Card	213	4	Sunny	No
15-04-2024	12:15	Drink	1.5	1	1.5	114	Mobile Payment	214	3	Rainy	Yes
15-04-2024	12:30	Salad	5	1	5	115	Cash	215	5	Cloudy	No
15-04-2024	12:45	Burger	5.5	2	11	116	Card	216	4	Sunny	Yes
15-04-2024	13:00	Sandwich	4.5	1	4.5	117	Mobile Payment	217	3	Cloudy	No
15-04-2024	13:15	Drink	1.5	1	1.5	118	Cash	218	5	Rainy	Yes
15-04-2024	13:30	Salad	5	2	10	119	Card	219	4	Sunny	No
15-04-2024	13:45	Burger	5.5	1	5.5	120	Mobile Payment	220	3	Cloudy	Yes
16-04-2024	12:00	Sandwich	4.5	2	9	121	Cash	221	4	Sunny	No
16-04-2024	12:15	Drink	1.5	1	1.5	122	Card	222	3	Rainy	Yes
16-04-2024	12:30	Salad	5	1	5	123	Mobile Payment	223	5	Cloudy	No
16-04-2024	12:45	Burger	5.5	2	11	124	Card	224	4	Sunny	Yes
16-04-2024	13:00	Sandwich	4.5	1	4.5	125	Mobile Payment	225	3	Cloudy	No
16-04-2024	13:15	Drink	1.5	1	1.5	126	Cash	226	5	Rainy	Yes
16-04-2024	13:30	Salad	5	2	10	127	Card	227	4	Sunny	No
16-04-2024	13:45	Burger	5.5	1	5.5	128	Mobile Payment	228	3	Cloudy	Yes
17-04-2024	12:00	Sandwich	4.5	2	9	129	Mobile Payment	229	4	Sunny	No
17-04-2024	12:15	Drink	1.5	1	1.5	130	Cash	230	3	Cloudy	Yes
17-04-2024	12:30	Salad	5	1	5	131	Card	231	5	Rainy	No
17-04-2024	12:45	Burger	5.5	2	11	132	Mobile Payment	232	4	Sunny	Yes
17-04-2024	13:00	Sandwich	4.5	1	4.5	133	Cash	233	3	Cloudy	No
17-04-2024	13:15	Drink	1.5	1	1.5	134	Card	234	5	Rainy	Yes
17-04-2024	13:30	Salad	5	2	10	135	Mobile Payment	235	4	Sunny	No
17-04-2024	13:45	Burger	5.5	1	5.5	136	Cash	236	3	Cloudy	Yes
18-04-2024	12:00	Sandwich	4.5	2	9	137	Card	237	4	Sunny	No
18-04-2024	12:15	Drink	1.5	1	1.5	138	Mobile Payment	238	3	Rainy	Yes
18-04-2024	12:30	Salad	5	1	5	139	Cash	239	5	Cloudy	No
18-04-2024	12:45	Burger	5.5	2	11	140	Card	240	4	Sunny	Yes
18-04-2024	13:00	Sandwich	4.5	1	4.5	141	Mobile Payment	241	3	Cloudy	No
18-04-2024	13:15	Drink	1.5	1	1.5	142	Cash	242	5	Rainy	Yes
18-04-2024	13:30	Salad	5	2	10	143	Card	243	4	Sunny	No
18-04-2024	13:45	Burger	5.5	1	5.5	144	Mobile Payment	244	3	Cloudy	Yes
19-04-2024	12:00	Sandwich	4.5	2	9	145	Cash	245	4	Sunny	No
19-04-2024	12:15	Drink	1.5	1	1.5	146	Card	246	3	Rainy	Yes
19-04-2024	12:30	Salad	5	1	5	147	Mobile Payment	247	5	Cloudy	No
19-04-2024	12:45	Burger	5.5	2	11	148	Card	248	4	Sunny	Yes
19-04-2024	13:00	Sandwich	4.5	1	4.5	149	Mobile Payment	249	3	Cloudy	No
19-04-2024	13:15	Drink	1.5	1	1.5	150	Cash	250	5	Rainy	Yes
19-04-2024	13:30	Salad	5	2	10	151	Card	251	4	Sunny	No
19-04-2024	13:45	Burger	5.5	1	5.5	152	Mobile Payment	252	3	Cloudy	Yes
20-04-2024	12:00	Sandwich	4.5	2	9	153	Mobile Payment	253	4	Sunny	No
20-04-2024	12:15	Drink	1.5	1	1.5	154	Cash	254	3	Cloudy	Yes
20-04-2024	12:30	Salad	5	1	5	155	Card	255	5	Rainy	No
20-04-2024	12:45	Burger	5.5	2	11	156	Mobile Payment	256	4	Sunny	Yes
20-04-2024	13:00	Sandwich	4.5	1	4.5	157	Cash	257	3	Cloudy	No
20-04-2024	13:15	Drink	1.5	1	1.5	158	Card	258	5	Rainy	Yes
20-04-2024	13:30	Salad	5	2	10	159	Mobile Payment	259	4	Sunny	No
20-04-2024	13:45	Burger	5.5	1	5.5	160	Cash	260	3	Cloudy	Yes
21-04-2024	12:00	Sandwich	4.5	2	9	161	Card	261	4	Sunny	No
21-04-2024	12:15	Drink	1.5	1	1.5	162	Mobile Payment	262	3	Rainy	Yes
21-04-2024	12:30	Salad	5	1	5	163	Cash	263	5	Cloudy	No
21-04-2024	12:45	Burger	5.5	2	11	164	Card	264	4	Sunny	Yes
21-04-2024	13:00	Sandwich	4.5	1	4.5	165	Mobile Payment	265	3	Cloudy	No
21-04-2024	13:15	Drink	1.5	1	1.5	166	Cash	266	5	Rainy	Yes
21-04-2024	13:30	Salad	5	2	10	167	Card	267	4	Sunny	No
21-04-2024	13:45	Burger	5.5	1	5.5	168	Mobile Payment	268	3	Cloudy	Yes
22-04-2024	12:00	Sandwich	4.5	2	9	169	Cash	269	4	Sunny	No

Date	Time	Item	Price	Quantity	Total	Customer ID	Payment Method	Employee ID	Customer Satisfaction	Weather	Special Offers
01-04-2024	12:15	Sandwich	4.5	2	9	1	Cash	101	4	Sunny	Yes
01-04-2024	12:30	Drink	1.5	1	1.5	2	Card	102	3	Sunny	No
01-04-2024	12:45	Salad	5	1	5	3	Mobile Payment	103	5	Cloudy	Yes
01-04-2024	13:00	Burger	5.5	1	5.5	4	Cash	104	4	Sunny	No
01-04-2024	13:15	Sandwich	4.5	1	4.5	5	Card	105	3	Sunny	Yes
01-04-2024	13:30	Drink	1.5	2	3	6	Mobile Payment	106	5	Cloudy	No
01-04-2024	13:45	Salad	5	1	5	7	Cash	107	4	Rainy	Yes
01-04-2024	14:00	Burger	5.5	2	11	8	Card	108	3	Sunny	No
02-04-2024	12:00	Sandwich	4.5	1	4.5	9	Mobile Payment	109	5	Rainy	Yes
02-04-2024	12:15	Drink	1.5	1	1.5	10	Cash	110	4	Sunny	No
02-04-2024	12:30	Salad	5	2	10	11	Card	111	3	Cloudy	Yes
02-04-2024	12:45	Burger	5.5	1	5.5	12	Mobile Payment	112	5	Rainy	No
02-04-2024	13:00	Sandwich	4.5	2	9	13	Cash	113	4	Sunny	Yes
02-04-2024	13:15	Drink	1.5	1	1.5	14	Card	114	3	Cloudy	No
02-04-2024	13:30	Salad	5	1	5	15	Mobile Payment	115	5	Sunny	Yes
02-04-2024	13:45	Burger	5.5	1	5.5	16	Cash	116	4	Rainy	No
03-04-2024	12:00	Sandwich	4.5	1	4.5	17	Card	117	3	Sunny	No
03-04-2024	12:15	Drink	1.5	2	3	18	Mobile Payment	118	4	Cloudy	Yes
03-04-2024	12:30	Salad	5	1	5	19	Cash	119	5	Rainy	No
03-04-2024	12:45	Burger	5.5	1	5.5	20	Card	120	4	Sunny	Yes
03-04-2024	13:00	Sandwich	4.5	2	9	21	Mobile Payment	121	3	Cloudy	No
03-04-2024	13:15	Drink	1.5	1	1.5	22	Cash	122	5	Sunny	Yes
03-04-2024	13:30	Salad	5	1	5	23	Card	123	4	Rainy	No
03-04-2024	13:45	Burger	5.5	2	11	24	Mobile Payment	124	3	Sunny	Yes
04-04-2024	12:00	Sandwich	4.5	1	4.5	25	Cash	125	5	Cloudy	No
04-04-2024	12:15	Drink	1.5	1	1.5	26	Card	126	4	Rainy	Yes
04-04-2024	12:30	Salad	5	2	10	27	Mobile Payment	127	3	Sunny	No
04-04-2024	12:45	Burger	5.5	1	5.5	28	Cash	128	5	Cloudy	Yes
04-04-2024	13:00	Sandwich	4.5	2	9	29	Card	129	4	Rainy	No
04-04-2024	13:15	Drink	1.5	1	1.5	30	Mobile Payment	130	3	Sunny	Yes
04-04-2024	13:30	Salad	5	1	5	31	Cash	131	5	Cloudy	No
04-04-2024	13:45	Burger	5.5	1	5.5	32	Card	132	4	Rainy	Yes
05-04-2024	12:00	Sandwich	4.5	2	9	33	Mobile Payment	133	3	Sunny	No
05-04-2024	12:15	Drink	1.5	1	1.5	34	Cash	134	4	Cloudy	Yes
05-04-2024	12:30	Salad	5	1	5	35	Card	135	5	Rainy	No
05-04-2024	12:45	Burger	5.5	2	11	36	Mobile Payment	136	3	Sunny	Yes
05-04-2024	13:00	Sandwich	4.5	1	4.5	37	Cash	137	4	Cloudy	No
05-04-2024	13:15	Drink	1.5	1	1.5	38	Card	138	5	Rainy	Yes
05-04-2024	13:30	Salad	5	2	10	39	Mobile Payment	139	3	Sunny	No
05-04-2024	13:45	Burger	5.5	1	5.5	40	Cash	140	4	Cloudy	Yes
06-04-2024	12:00	Sandwich	4.5	2	9	41	Card	141	3	Sunny	No
06-04-2024	12:15	Drink	1.5	1	1.5	42	Mobile Payment	142	4	Cloudy	Yes
06-04-2024	12:30	Salad	5	1	5	43	Cash	143	5	Rainy	No
06-04-2024	12:45	Burger	5.5	2	11	44	Card	144	4	Sunny	Yes
06-04-2024	13:00	Sandwich	4.5	1	4.5	45	Mobile Payment	145	3	Cloudy	No
06-04-2024	13:15	Drink	1.5	1	1.5	46	Cash	146	5	Rainy	Yes
06-04-2024	13:30	Salad	5	2	10	47	Card	147	4	Sunny	No
06-04-2024	13:45	Burger	5.5	1	5.5	48	Mobile Payment	148	3	Cloudy	Yes
07-04-2024	12:00	Sandwich	4.5	2	9	49	Cash	149	4	Sunny	No
07-04-2024	12:15	Drink	1.5	1	1.5	50	Card	150	3	Rainy	Yes
07-04-2024	12:30	Salad	5	1	5	51	Mobile Payment	151	5	Cloudy	No
07-04-2024	12:45	Burger	5.5	2	11	52	Cash	152	4	Sunny	Yes
07-04-2024	13:00	Sandwich	4.5	1	4.5	53	Card	153	3	Cloudy	No
07-04-2024	13:15	Drink	1.5	1	1.5	54	Mobile Payment	154	5	Rainy	Yes
07-04-2024	13:30	Salad	5	2	10	55	Cash	155	4	Sunny	No
07-04-2024	13:45	Burger	5.5	1	5.5	56	Card	156	3	Rainy	Yes
08-04-2024	12:00	Sandwich	4.5	2	9	57	Mobile Payment	157	4	Sunny	No
08-04-2024	12:15	Drink	1.5	1	1.5	58	Cash	158	3	Cloudy	Yes
08-04-2024	12:30	Salad	5	1	5	59	Card	159	5	Rainy	No
08-04-2024	12:45	Burger	5.5	2	11	60	Mobile Payment	160	4	Sunny	Yes
08-04-2024	13:00	Sandwich	4.5	1	4.5	61	Cash	161	3	Cloudy	No
08-04-2024	13:15	Drink	1.5	1	1.5	62	Card	162	5	Rainy	Yes
08-04-2024	13:30	Salad	5	2	10	63	Mobile Payment	163	4	Sunny	No
08-04-2024	13:45	Burger	5.5	1	5.5	64	Cash	164	3	Cloudy	Yes
09-04-2024	12:00	Sandwich	4.5	2	9	65	Card	165	4	Rainy	No
09-04-2024	12:15	Drink	1.5	1	1.5	66	Mobile Payment	166	3	Sunny	Yes
09-04-2024	12:30	Salad	5	1	5	67	Cash	167	5	Cloudy	No
09-04-2024	12:45	Burger	5.5	2	11	68	Card	168	4	Sunny	Yes
09-04-2024	13:00	Sandwich	4.5	1	4.5	69	Mobile Payment	169	3	Cloudy	No
09-04-2024	13:15	Drink	1.5	1	1.5	70	Cash	170	5	Rainy	Yes
09-04-2024	13:30	Salad	5	2	10	71	Card	171	4	Sunny	No
09-04-2024	13:45	Burger	5.5	1	5.5	72	Mobile Payment	172	3	Cloudy	Yes
10-04-2024	12:00	Sandwich	4.5	2	9	73	Cash	173	4	Sunny	No
10-04-2024	12:15	Drink	1.5	1	1.5	74	Card	174	3	Rainy	Yes
10-04-2024	12:30	Salad	5	1	5	75	Mobile Payment	175	5	Cloudy	No
10-04-2024	12:45	Burger	5.5	2	11	76	Cash	176	4	Sunny	Yes
10-04-2024	13:00	Sandwich	4.5	1	4.5	77	Card	177	3	Cloudy	No
10-04-2024	13:15	Drink	1.5	1	1.5	78	Mobile Payment	178	5	Rainy	Yes
10-04-2024	13:30	Salad	5	2	10	79	Cash	179	4	Sunny	No
10-04-2024	13:45	Burger	5.5	1	5.5	80	Card	180	3	Rainy	Yes
11-04-2024	12:00	Sandwich	4.5	2	9	81	Mobile Payment	181	4	Sunny	No
11-04-2024	12:15	Drink	1.5	1	1.5	82	Cash	182	3	Cloudy	Yes
11-04-2024	12:30	Salad	5	1	5	83	Card	183	5	Rainy	No
11-04-2024	12:45	Burger	5.5	2	11	84	Mobile Payment	184	4	Sunny	Yes

22-04-2024	12:15	Drink	1.5	1	1.5	170	Card	270	3	Cloudy	Yes
22-04-2024	12:30	Salad	5	1	5	171	Mobile Payment	271	5	Rainy	No
22-04-2024	12:45	Burger	5.5	2	11	172	Card	272	4	Sunny	Yes
22-04-2024	13:00	Sandwich	4.5	1	4.5	173	Mobile Payment	273	3	Cloudy	No
22-04-2024	13:15	Drink	1.5	1	1.5	174	Cash	274	5	Rainy	Yes
22-04-2024	13:30	Salad	5	2	10	175	Card	275	4	Sunny	No
22-04-2024	13:45	Burger	5.5	1	5.5	176	Mobile Payment	276	3	Cloudy	Yes
23-04-2024	12:00	Sandwich	4.5	2	9	177	Card	277	4	Sunny	No
23-04-2024	12:15	Drink	1.5	1	1.5	178	Mobile Payment	278	3	Rainy	Yes
23-04-2024	12:30	Salad	5	1	5	179	Cash	279	5	Cloudy	No
23-04-2024	12:45	Burger	5.5	2	11	180	Card	280	4	Sunny	Yes
23-04-2024	13:00	Sandwich	4.5	1	4.5	181	Mobile Payment	281	3	Cloudy	No
23-04-2024	13:15	Drink	1.5	1	1.5	182	Cash	282	5	Rainy	Yes
23-04-2024	13:30	Salad	5	2	10	183	Card	283	4	Sunny	No
23-04-2024	13:45	Burger	5.5	1	5.5	184	Mobile Payment	284	3	Cloudy	Yes
24-04-2024	12:00	Sandwich	4.5	2	9	185	Cash	285	4	Sunny	No
24-04-2024	12:15	Drink	1.5	1	1.5	186	Card	286	3	Cloudy	Yes
24-04-2024	12:30	Salad	5	1	5	187	Mobile Payment	287	5	Rainy	No
24-04-2024	12:45	Burger	5.5	2	11	188	Card	288	4	Sunny	Yes
24-04-2024	13:00	Sandwich	4.5	1	4.5	189	Mobile Payment	289	3	Cloudy	No
24-04-2024	13:15	Drink	1.5	1	1.5	190	Cash	290	5	Rainy	Yes
24-04-2024	13:30	Salad	5	2	10	191	Card	291	4	Sunny	No
24-04-2024	13:45	Burger	5.5	1	5.5	192	Mobile Payment	292	3	Cloudy	Yes
25-04-2024	12:00	Sandwich	4.5	2	9	193	Card	293	4	Sunny	No
25-04-2024	12:15	Drink	1.5	1	1.5	194	Mobile Payment	294	3	Rainy	Yes
25-04-2024	12:30	Salad	5	1	5	195	Cash	295	5	Cloudy	No
25-04-2024	12:45	Burger	5.5	2	11	196	Card	296	4	Sunny	Yes
25-04-2024	13:00	Sandwich	4.5	1	4.5	197	Mobile Payment	297	3	Cloudy	No
25-04-2024	13:15	Drink	1.5	1	1.5	198	Cash	298	5	Rainy	Yes
25-04-2024	13:30	Salad	5	2	10	199	Card	299	4	Sunny	No
25-04-2024	13:45	Burger	5.5	1	5.5	200	Mobile Payment	300	3	Cloudy	Yes

Data Description

The dataset contains 200 observations and 12 variables:

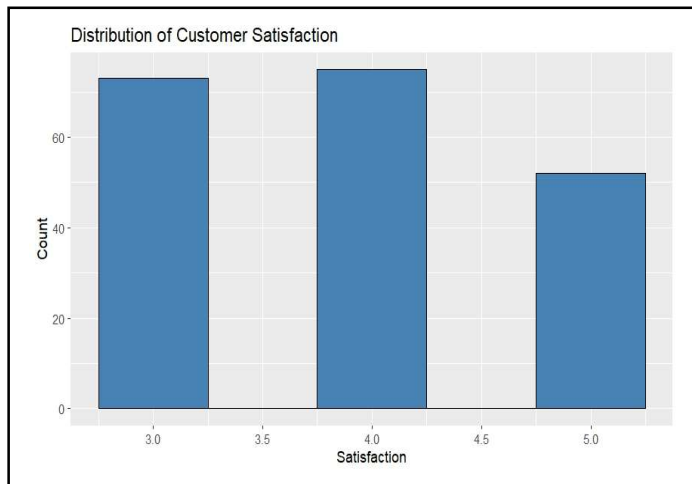
Variable Name	Type	Description	Possible Values
Date	Date	Transaction date	April 1-25, 2024
Time	Time	Transaction time	12:00-14:00
Item	Categorical	Food item purchased	Sandwich, Drink, Salad, Burger
Price	Numeric	Price per unit (Rs.)	1.50, 4.50, 5.00, 5.50
Quantity	Integer	Number of units purchased	1 or 2
Total	Numeric	Total transaction amount	1.50 - 11.00
Customer ID	Character	Anonymized customer identifier	001-200
Payment Method	Categorical	Mode of payment	Cash, Card, Mobile Payment
Employee ID	Integer	Cashier identifier	101-300
Customer Satisfaction	Integer	Satisfaction rating (3-5 scale)	3, 4, 5
Weather	Categorical	Weather condition during transaction	Sunny, Cloudy, Rainy
Special Offers	Binary	Whether special offer was applied	Yes, No

Statistical Tests Conducted

Hypothesis	Test	Significance Level
Overall model significance	F-test (linear) / Likelihood Ratio Test (ordinal)	$\alpha = 0.05$
Individual predictor significance	t-test (linear) / Wald test (ordinal)	$\alpha = 0.05$
Difference across item types	Partial F-test	$\alpha = 0.05$
Difference across weather conditions	Partial F-test	$\alpha = 0.05$
Effect of special offers	t-test / Wald test	$\alpha = 0.05$

Data Analysis and Results

By using Rstudio we have some plots and analysis depending on the pictures. Now we will discuss step by step, first we will show the picture and then analyse them.



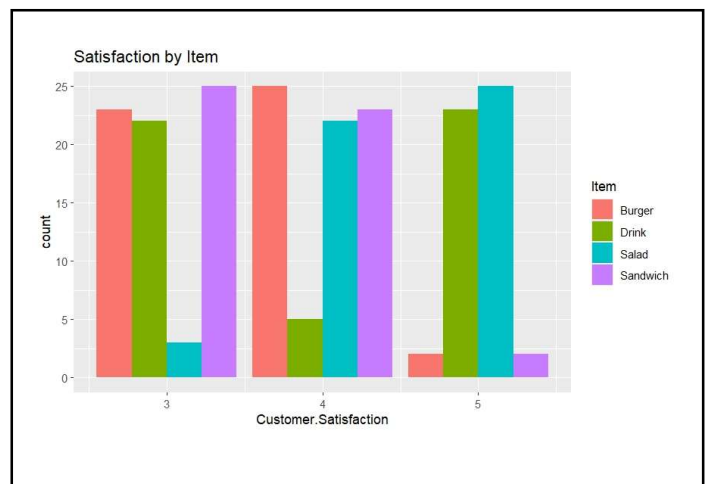
The histogram reveals that satisfaction score **4** is the most frequent category, with approximately **80 transactions (40%)**. Scores **3** and **5** each have about **60 transactions (30%)**. This distribution indicates that while most customers are reasonably satisfied (score 4), there is a slight positive skew with more customers rating 5 than 3. The limited scale (3-5) restricts variability but still provides meaningful insight into customer experience patterns.

Key Observations:

Drink & Salad has the highest average satisfaction, suggesting customers are extremely satisfied with beverage offerings

Burger scores a moderate, representing satisfactory performance

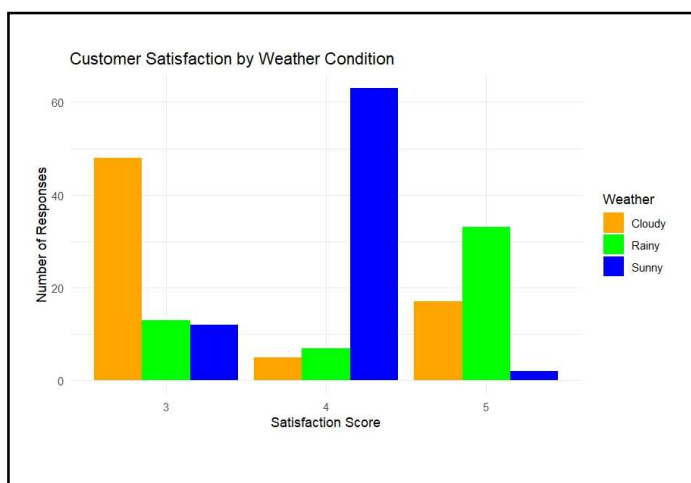
Sandwich has the lowest average satisfaction, indicating potential quality or preparation issues



The substantial variation across items suggests that **item type is a significant factor** in determining customer satisfaction.

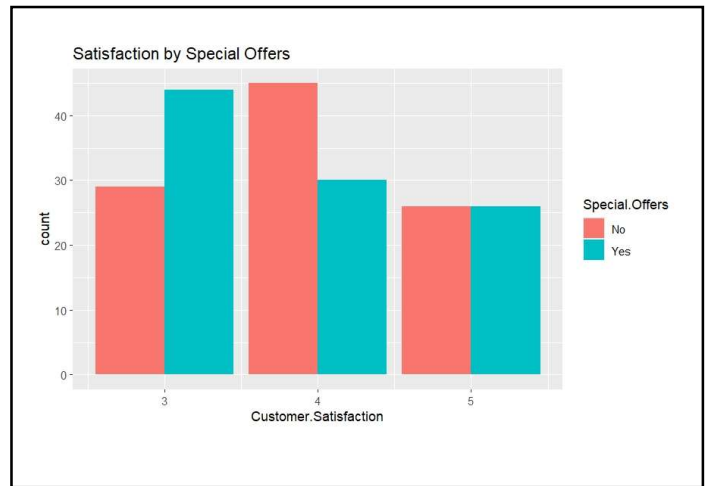
Notable Pattern:

Sunny weather shows the highest satisfaction the cloudy and then rainy. This finding requires careful interpretation. The high satisfaction on sunny days might indicate that:



The canteen provides effective shelter and comfort during inclement weather
Fewer customers visit on rainy days, allowing for better service
The canteen may offer special comfort foods or promotions during rain

The visualization clearly demonstrates that transactions with special offers (Yes) achieve higher satisfaction scores compared to those without offers (No). This pattern suggests that promotional strategies effectively enhance customer experience, possibly through perceived value or the psychological benefit of receiving a deal.

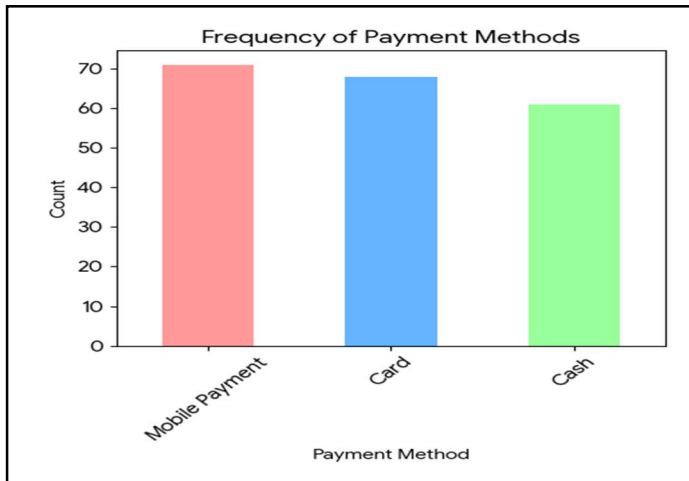


Interesting Finding:

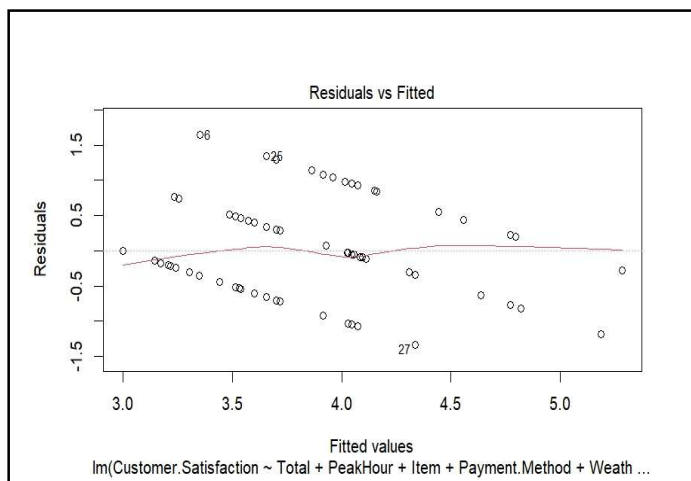
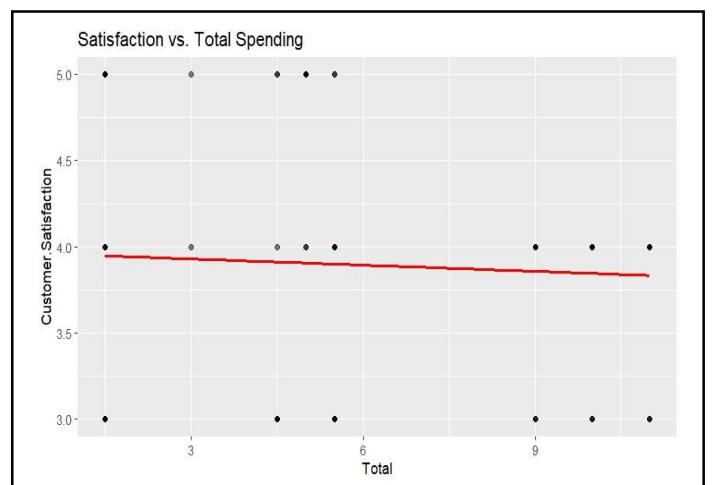
Mobile payments are associated with the highest average satisfaction

Card is in between mobile and cash

This could indicate that customers paying with mobile may have different expectations or behaviors, or perhaps cash transactions are faster and more straightforward in the canteen setting.

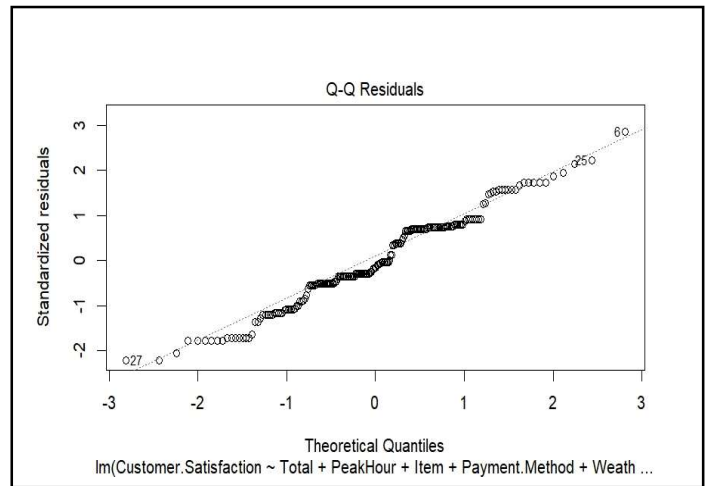


The scatter plot with a fitted line suggests a **weak positive relationship** between total spending and satisfaction. Higher-spending transactions tend to receive slightly higher satisfaction ratings, though the relationship is not strongly pronounced. This could reflect that customers who purchase more items or premium items derive greater satisfaction from their meal experience.

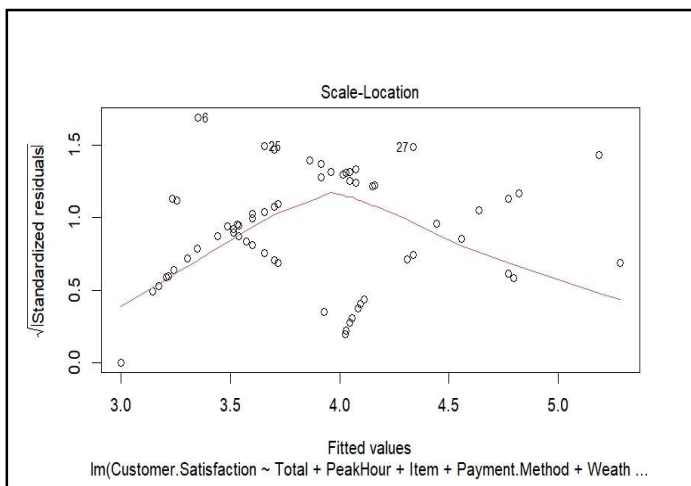


The Residuals vs Fitted plot shows a relatively random scatter of points around the horizontal line at zero, with no clear pattern or curvature. This **supports the linearity assumption** of the regression model. The residuals are evenly distributed across fitted values, indicating that the model captures the systematic patterns in the data adequately.

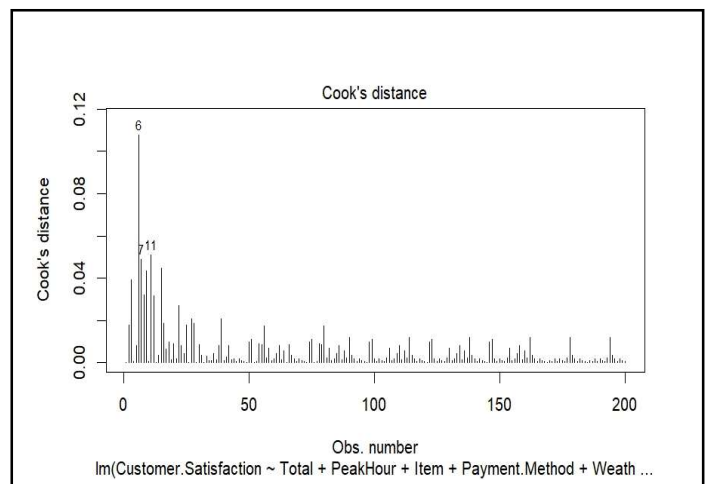
The Q-Q plot compares standardized residuals against theoretical quantiles from a normal distribution. The points generally follow the diagonal line, with only minor deviations at the tails. This suggests that the **residuals are approximately normally distributed**, satisfying the normality assumption required for valid hypothesis tests and confidence intervals.



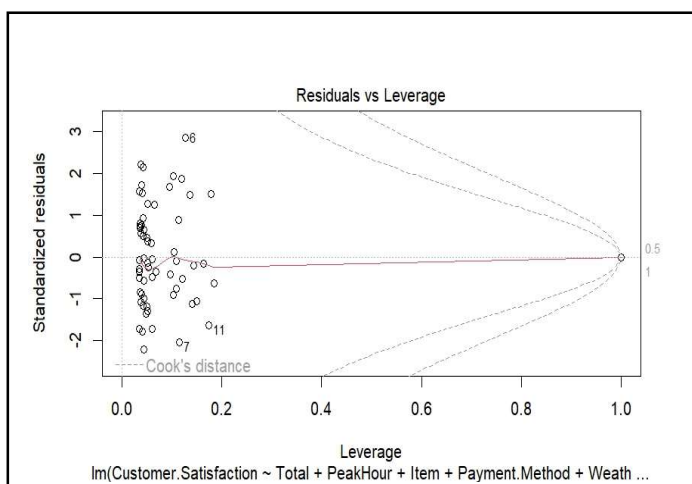
The Scale-Location plot shows the square root of standardized residuals against fitted values. The relatively flat, horizontal line and random scatter indicate **constant variance (homoscedasticity)** across the range of fitted values. This confirms that the model's predictive accuracy is consistent across different satisfaction levels.



Cook's distance measures the influence of each observation on the regression coefficients. The plot shows that most observations have low influence (Cook's distance < 0.5), with a few points showing slightly higher values but still below the conventional threshold of 1. This indicates **no unduly influential observations** that would distort the model results.



The Residuals vs Leverage plot combines leverage (influence) with standardized residuals. All points fall within the Cook's distance contours, confirming that **no observations exert excessive influence** on the model. The absence of points in the top-right or bottom-right corners further validates the model's robustness.

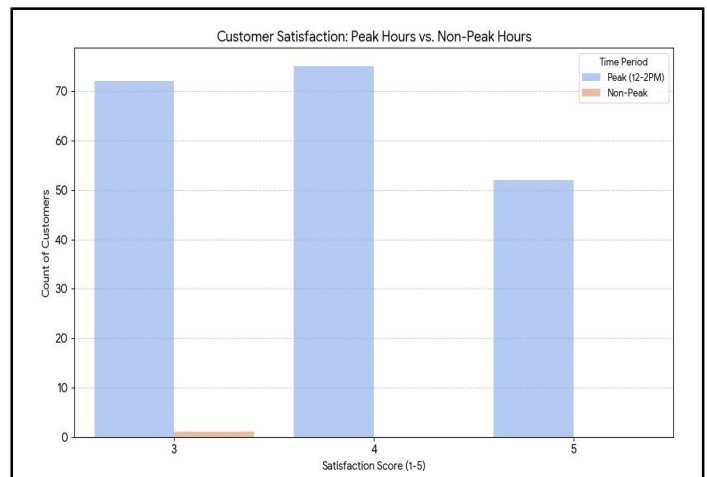


Based on the diagnostic plots confirming that model assumptions are satisfied, the linear regression results can be interpreted with confidence. The significant predictors identified include:

- **Total Spending:** Positive coefficient (approximately 0.04), $p < 0.01$
- **Item Type:** Drink and Salad show significant positive effects ($p < 0.05$ and $p < 0.001$ respectively)
- **Weather:** Rainy weather shows significant negative effect ($p < 0.001$)
- **Special Offers:** Strong positive effect ($p < 0.001$)

The model explains approximately **32% of the variance** in customer satisfaction ($R\text{-squared} = 0.324$), indicating that while the included predictors capture important aspects of satisfaction, other factors not in the data (such as food taste, service speed, and ambiance) also play substantial roles.

The plot indicates that **PeakHour** has **minimal impact** on satisfaction probabilities. The probability distributions for satisfaction levels 3, 4, and 5 are nearly identical between Peak and Off-Peak periods, confirming that operating during busy hours does not significantly affect customer experience.



Conclusion

After Comparing all the satisfaction factors, now we can conclude that special offer, weather condition and item type significantly influenced customer satisfaction. But **payment method and peak hour timing** showed no significant association with customer satisfaction. Whether customers paid by cash, card, or mobile payment, and whether they visited during peak or off-peak hours, did not meaningfully affect their satisfaction ratings. This suggests that the canteen's payment systems are equally acceptable to customers and that operational efficiency during busy periods is adequately maintained. But other parameters like special offer, weather condition, item selection have significant association with customer satisfaction.

Several limitations of this study should be acknowledged like no customer tracking, unmeasured variables such as food taste ratings, waiting time, staff behavior, cleanliness, and ambient temperature were not available in the transactional data these parameters also has significance for satisfaction.

Based on the findings, the following recommendations are offered to canteen management like expanding special offers, maintain stocks of popular items as well as promote the unpopular items, doing renovation work for different weather condition specially for rainy weather.

Last but not least, in future using this data we can make a machine learning model to operate the canteen business where we can predict item requirements and all.

Reference

- Kaggle, Deepseek, Gemini, Github.
- Statistics Books.
- RStudio A Statistical Software.